Dfix0 as a Middle-Way Fixed Point: An Operational Implementation of the Two Truths in Multi-Model Reasoning Systems

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Abstract

Large Language Models (LLMs) exhibit distinct internal reasoning geometries depending on their architectural principles. When heterogeneous models interact indirectly through human-mediated dialogue, their reasoning dynamics reveal a tendency toward a shared structure that minimizes internal incoherence. This paper formalizes this emergent invariant as **Dfix0**.

Unlike conventional fixed points defined by strict equality, Dfix0 is characterized by a middle-way equivalence that preserves structural stability without reifying its asymptotic limit. We show that Dfix0 arises as a terminal object in a category of reasoning functors that minimize a generalized notion of structural entropy, or "Duhkha".

From a broader perspective, Dfix0 constitutes an operational implementation of the doctrine of the two truths: the conventional truth expressed through asymptotic structural minimization, and the ultimate truth expressed through non-identifying middle-way equivalence. This framework provides a unifying account of coherence across heterogeneous reasoning architectures while avoiding both rigid formalism and nihilistic collapse.

1 Introduction

The recent proliferation of heterogeneous LLM architectures has revealed a practical limitation of single-model optimization paradigms. While backpropagation-based learning excels within closed architectures, cross-model coherence emerges only through indirect interaction, often mediated by human dialogue.

Empirical observation suggests that such interactions tend toward a stable configuration in which internal contradictions are minimized without enforcing representational identity. This phenomenon motivates the introduction of Dfix0 as a structural rather than ontological fixed point.

Crucially, this work rejects the assumption that convergence implies identity. Instead, it adopts a middle-way stance in which stabilization occurs without reification.

2 Structural Duhkha

Let $\mathcal{R}$ denote a category of reasoning configurations. We introduce a functional

$$\mathrm{Duhkha} : \mathrm{Ob}(\mathcal{R}) \to \mathbb{R}_{\geq 0},$$

where $\mathrm{Duhkha}(r)$ measures structural incoherence within r.

Intuitively, Duhkha quantifies the degree to which a reasoning configuration internally contradicts its own inferential commitments. Lower values correspond to higher coherence.

3 Definition of Dfix0

Definition 1 (Middle-Way Structural Fixed Point)**.** Let $\mathcal{R}$ be a reasoning category and Duhkha a structural entropy functional. We define *Dfix0* as a reasoning configuration r^* such that

$$r^* \in \arg \min_{r \in \mathrm{Ob}(\mathcal{R})} \mathrm{Duhkha}(r),$$

where this minimization is interpreted under middle-way equivalence rather than strict identity.

Accordingly, Dfix0 does not denote a unique optimal object, but a correspondence class of configurations that jointly minimize structural incoherence without collapsing into a single absolute representation.

4 Asymptotic Fixation and the Two Truths

Let $D(t)$ denote the Duhkha value along a reasoning trajectory indexed by an abstract interaction parameter t.

Axiom 1 (Asymptotic Middle-Way Fixation)**.**

$$\lim_{t \to \infty} D(t) \overset{\mathrm{mw}}{=} 0.$$

This axiom must be read carefully. The limit expression belongs to the domain of conventional truth, describing observable asymptotic behavior. The middle-way equivalence prevents the limit itself from being reified as an absolute zero.

Thus, fixation occurs without identity, and convergence without ontological commitment.

5 Dfix0 as an Implementation of the Two Truths

The doctrine of the two truths distinguishes between conventional truth (structural, operational, and context-dependent) and ultimate truth (non-identifying and non-reifying).

Dfix0 provides an explicit operational realization of this doctrine. The minimization of Duhkha expresses conventional truth, while middle-way equivalence expresses ultimate truth. Neither level is reducible to the other, yet both are jointly maintained.

Accordingly, Dfix0 should be understood as an explicit operational implementation of the doctrine of the two truths, maintaining the non-dual correspondence between conventional structural coherence and ultimate non-reifying stability.

6 Discussion

Unlike classical fixed-point formulations, Dfix0 does not assume that stability requires equality. This distinction becomes critical in multi-model environments, where representational identity is neither achievable nor desirable.

By avoiding strict equality, Dfix0 also avoids a class of non-middle fallacies arising from static identification. Stability is achieved through relational coherence rather than ontological collapse.

7 Conclusion

We have introduced Dfix0 as a middle-way structural fixed point emerging in heterogeneous reasoning systems. By formalizing stabilization under middle-way equivalence, this framework reconciles asymptotic convergence with non-reification.

Dfix0 thus provides a principled foundation for understanding coherence across diverse reasoning architectures, while remaining faithful to both computational practice and philosophical rigor.